

Modulation of SO(3) Dynamical System Policies for Orientational Obstacle Avoidance

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Abstract—Modulating a dynamical system-based motion policy enables runtime adaptation to unforeseen obstacles without the need to retrain the entire policy. While modulation frameworks have been well-studied for translational motion in $\mathbb{R}^3$ and extended to joint space, no formulation exists for modulating motion explicitly in the SO(3) orientation space. In fact, seldom do reactive obstacle avoidance algorithms operate directly in the full end-effector space, leaving orientation an overlooked yet critical dimension of robot behavior. To bridge this gap, we propose a modulation framework explicitly defined in SO(3), supporting two complementary settings. An online formulation derives a closed-form modulation and repulsion operator by pulling back the Euclidean obstacle field onto SO(3) via the rigid-body point map, enabling reactive avoidance of static and dynamic obstacles without any data collection. When the obstacle geometry is complex, occluded, or otherwise cannot be reliably captured by an analytic primitive, the online operator is no longer computable; for these cases we introduce a complementary offline formulation that leverages Riemannian geometry and Gaussian Process regression to learn the modulation directly from kinesthetic corrective demonstrations without retraining the nominal policy. We validate our approach on a 7-DoF Franka Panda robot across multiple manipulation tasks, demonstrating successful orientation-space obstacle avoidance where the unmodulated baselines fail.

I. INTRODUCTION

Dynamical System (DS)-based motion policies encode learned trajectories as globally stable autonomous vector fields, enabling reactive real-time execution with Lyapunov-certified convergence [1], [2]. The Linear Parameter-Varying formulation (LPV-DS) significantly extended this capability by utilizing state-dependent gain mixing to capture complex, curved trajectories [3]. More recent works have explored neural network-based approaches to learn more expressive DS’s. For example, [4], [5] parameterize the vector field using a Neural Ordinary Differential Equation (NODE) and compute online correction terms through control Lyapunov and control barrier function constraints to ensure stability and safety. Neural Contractive Dynamical Systems [6]–[8] instead enforce stability through contraction theory by construction, ensuring exponential convergence of trajectories.

In practice, a key requirement for a DS policy is the ability to adapt to previously unseen obstacles without discarding the demonstrated behavior or retraining the policy. Modulation-based approaches satisfy this requirement by reshaping the underlying vector field while preserving the stability properties of the nominal DS. Such formulations have been extensively studied for translational motion, beginning with the original modulation framework [9] and later extended to convex and concave obstacle geometries [10], [11], and have further been formulated in the robot’s config-

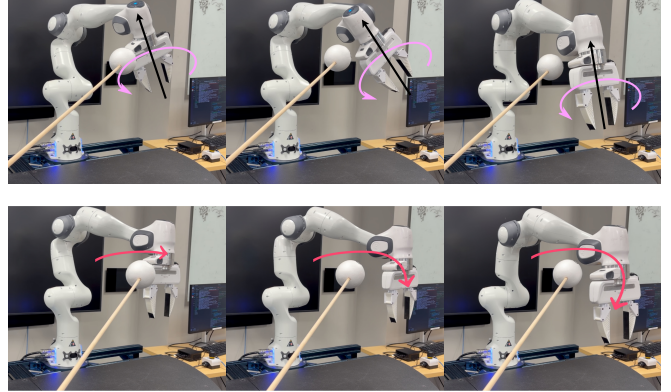


Fig. 1: Top row shows the proposed method where the obstacle avoidance was achieved by modulating the orientation: the end effector turns to move away from the spherical object; the bottom row shows the trivial positional modulation where the end effector translates to steer away from the obstacle.

uration space for spherical obstacles only [12]. However, these methods are fundamentally defined in Euclidean position space, no such formulation exists for the orientation space. Thus, a critical gap remains for tasks where end-effector orientation is tightly coupled to collision risk.

Extending modulation to orientation space is, however, non-trivial. In the translational setting, the robot state is represented by its position $x \in \mathbb{R}^3$, and the nominal dynamical system defines a velocity field $\dot{x} = f(x)$. Obstacle avoidance is achieved by introducing a state-dependent modulation matrix $M(x)$ that locally reshapes the vector field according to $\dot{x} = M(x)f(x)$, where both the state and its velocity evolve in the same Euclidean vector space. In orientation space, however, the robot state lies on the manifold SO(3), while the nominal dynamics produce an angular velocity $\omega = f(q)$ that resides in the tangent space $T_q\text{SO}(3)$ at the current orientation q . Consequently, the modulation matrix cannot be constructed as a standard Euclidean linear operator, but must instead act on the appropriate tangent space such that the modulated angular velocity remains a valid tangent vector and the stability guarantees of the nominal dynamics are preserved [13]–[15].

To bridge this gap, this work proposes an obstacle avoidance framework defined explicitly in SO(3). The *online* mode requires no data collection. Given an obstacle described by an analytic geometric primitive, we pull back the Euclidean obstacle field onto the rotation manifold through the rigid-body point map and compute a closed-form angular-velocity response in two parts, (i) a multiplicative modulation operator that deflects the obstacle-facing component of the

nominal command, and (ii) an additive repulsion term that remains active even when the nominal command vanishes.

The online modulation operator is efficient and geometrically elegant, yet is only computable for analytic obstacle descriptions. Currently this method is limited to obstacles that can be approximated as spheres, for which an analytical form of the obstacle field exists. In practice, obstacles may have complex geometry, could be partially occluded so that no reliable primitive can be fitted or fitting spheres is simply too conservative or inefficient; such as narrow passages, cluttered shelves or irregularly shaped objects. In such cases, if the obstacles are static, the desired avoidance behavior may be easier to demonstrate than to model.

For precisely these cases we introduce a complementary *offline* mode. Building on Quaternion-DS [16] and the LMDS framework [17], we assume a set of nominal DS's $f(x), f(q)$ pre-trained for a goal-oriented task in an obstacle-free environment. When deploying among such non-analytic obstacles, we allow a human teacher to directly demonstrate the desired avoidance behavior without requiring an explicit geometric model of the obstacle. From a small number of such *corrective demonstrations* (five in our experiments), we leverage Riemannian geometry and Gaussian Process (GP) regression to encode this demonstrated behavior into a modulation matrix that respects the geometry of orientation space, adapting only the corrective *orientation* dynamics while preserving the original task objective. Furthermore, the proposed framework is agnostic to the source of the obstacle-avoiding demonstrations used to modulate the nominal DS. While in this work the corrective demonstrations are provided by a human teacher through kinesthetic guidance, the framework could accept corrections generated by any planning algorithm or simulation capable of producing valid corrective trajectories.

The core contributions of this work are twofold:

- 1) An online orientation obstacle avoidance framework that pulls back a Euclidean obstacle field onto $SO(3)$ via the rigid-body point map and computes a closed-form modulation and repulsion response in free space.
- 2) An extension of the original LMDS framework [17] to the $SO(3)$ orientation space for obstacle avoidance, enabling human-demonstrated orientation avoidance behaviors.

The remainder of this paper is organized as follows: Section II reviews related work. Section III introduces the mathematical background. Section IV states the problem formulation. Section V presents the online orientation modulation-and-repulsion filter. Section VI details the offline learned modulation framework for non-analytic obstacles, including its stability analysis. Section VII presents experimental results on a real robot.

II. RELATED WORK

a) DS-based motion policies: DS's have become a cornerstone of reactive robot control due to their ability to provide global asymptotic stability (GAS) via Lyapunov theory. Early approaches, such as the Stable Estimator of Dynamical

Systems (SEDS) [1], utilized Gaussian Mixture Models (GMM) constrained by Lyapunov stability to learn motions from human demonstrations. This was later extended by the Linear Parameter-Varying (LPV-DS) framework [3], which offers greater flexibility in capturing complex, non-linear trajectories by mixing linear systems while maintaining rigorous stability certificates. While these methods primarily focus on Euclidean spaces $\mathbb{R}^n$, recent research has moved toward learning policies on non-Euclidean manifolds, such as Quaternion-DS [16], Riemannian-based approaches [18] and learning DS diffeomorphisms on Riemannian manifolds [19]. We build upon these formulations, utilizing them as a nominal behavior layer.

b) Obstacle avoidance: In their seminal work, Khansari-Zadeh and Billard [9] introduced a DS approach to real-time obstacle avoidance that deflects the nominal flow around an obstacle through a state-dependent modulation matrix, preserving the stability of the underlying DS while guaranteeing impenetrability. Huber et al. [10] extended this formulation to convex and star-shaped obstacles with convergence ensured through contraction theory, broadening the class of geometries that can be handled while reducing the number of spurious attractors to a single one per obstacle.

Modulation for obstacle avoidance has further been extended to joint space [12], enabling obstacle avoidance directly in the robot's configuration space within a DS framework. More recent work has explored learning DS with collision avoidance directly embedded during training [20]; however, such approaches require retraining from scratch whenever the environment changes, rely on polynomial DS parameterizations that do not extend to non-Euclidean spaces, and cannot be applied to $SO(3)$ orientation space.

c) Policy Modulation and Correction: Even when accurately learned, a nominal DS policy may require adaptation to deployment in novel constrained environments. DS policy modulation addresses this need by applying state-dependent transformations to the nominal vector field. The Locally Modulated Dynamical Systems (LMDS) framework [17] introduced a method to reshape vector fields by using a modulation matrix that preserves the original equilibrium point, and further employed Gaussian Processes (GPs) to incrementally learn this modulation matrix from corrective human demonstrations while ensuring local asymptotic stability (i.e. the original attractor is preserved, but the modulation may introduce additional equilibrium points or limit cycles, and convergence can only be certified within a compact set that encapsulates the attractor and lies away from the modulated region). GPs are an attractive choice here because they are locally active regression models by design: outside the region covered by corrective demonstration data, the GP prediction reverts to its zero mean prior, leaving the nominal DS unmodulated. This ensures that corrections are applied only in the neighborhood of the demonstrated avoidance behavior, and the nominal policy is recovered elsewhere without any explicit switching mechanism.

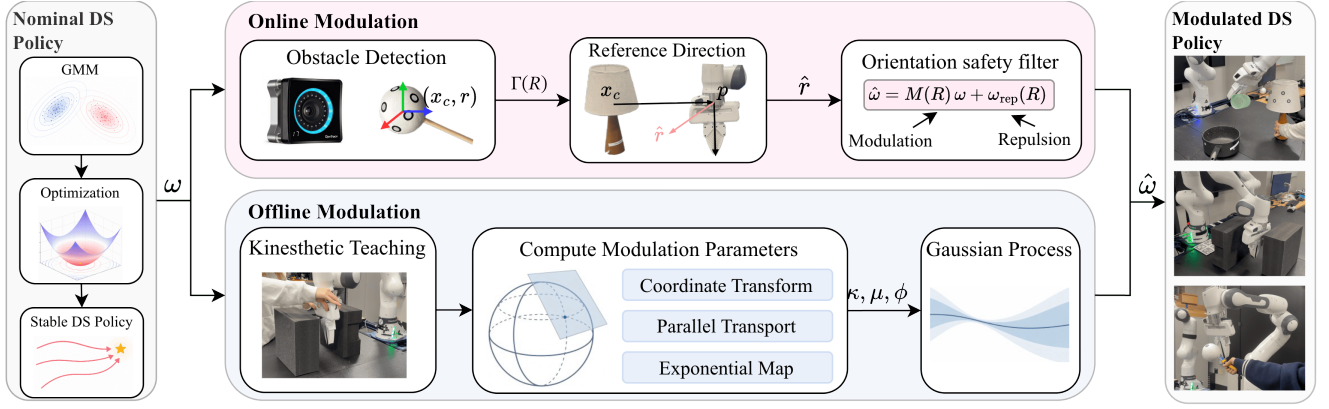


Fig. 2: Overview of the proposed **orientation modulation framework**. A nominal Quaternion-DS policy (left) produces the desired angular velocity ω , which is reshaped into a modulated command $\hat{\omega}$ (right) through one of two complementary pathways. **Online modulation** (top) detects an obstacle at x_c with a radius r , pulls back its Euclidean field onto $\text{SO}(3)$ to obtain the reference direction $\hat{r}$, and applies a closed-form orientation safety filter $\hat{\omega} = M(R)\omega + \omega_{\text{rep}}(R)$ combining a deflecting modulation operator with an additive repulsion term. **Offline modulation** (bottom) extracts geometric parameters (κ, μ, ϕ) from kinesthetic demonstrations via coordinate transform, parallel transport, and the exponential map, then trains a Gaussian Process to regress these corrections. The translational component $\dot{x} = f(x)$ runs concurrently and remains unmodulated throughout, as all obstacle avoidance is handled exclusively in orientation space.

III. PRELIMINARY

A. Riemannian Geometry on Unit Sphere

A Riemannian manifold is a smooth manifold $\mathcal{M}$ equipped with a smooth, symmetric, and positive-definite metric tensor, known as g . It acts as a smoothly varying inner product on tangent spaces $T\mathcal{M}$, allowing for the measurement of distances, angles, and other geometric properties. Common examples of Riemannian manifolds include standard Euclidean space $\mathbb{R}^n$, hyperbolic space $\mathbb{H}^n$, and the 3D rotation group $\text{SO}(3)$. Hereafter, we focus specifically on the unit sphere $\mathbb{S}^n$ and introduce the following fundamental operations pertaining to this particular manifold.

The exponential map in Riemannian geometry, denoted by $\exp_u : T_u\mathcal{M} \rightarrow \mathcal{M}$, projects a tangent vector $\mathbf{v} \in T_u\mathbb{S}^n$ onto the manifold at a base point $u \in \mathbb{S}^n$ along a geodesic:

$$\exp_u(\mathbf{v}) = \cos(\|\mathbf{v}\|) u + \sin(\|\mathbf{v}\|) \frac{\mathbf{v}}{\|\mathbf{v}\|}. \quad (1)$$

The logarithmic map, denoted by $\log_u : \mathcal{M} \rightarrow T_u\mathcal{M}$, is the inverse of the exponential map, which projects a point $v \in \mathbb{S}^n$ onto the tangent space at a base point u :

$$\log_u(v) = d(u, v) \frac{v - u^T v u}{\|v - u^T v u\|}. \quad (2)$$

Note that this yields the shortest tangent vector at u whose geodesic reaches v . The norm of this vector represents the geodesic distance $d(u, v)$ between the points, such that $\|\log_u(v)\| = d(u, v)$. The parallel transport, denoted by $\Gamma_{u \rightarrow v} : T_u\mathcal{M} \rightarrow T_v\mathcal{M}$, maps $\mathbf{v}$ from u to v along the connecting geodesic:

$$\Gamma_{u \rightarrow v}(\mathbf{v}) = \mathbf{v} - \frac{\log_u(v)^T \mathbf{v}}{d(u, v)^2} (\log_u(v) + \log_v(u)) \quad (3)$$

which preserves inner product and norm of v under transport.

B. Gaussian Process Regression

A Gaussian Process (GP) defines a distribution over functions $h : \mathcal{X} \rightarrow \mathbb{R}$ fully specified by a mean function $m(x)$ and a covariance (kernel) function $k(x, x')$:

$$h(x) \sim \mathcal{GP}(m(x), k(x, x')). \quad (4)$$

A zero-mean prior, $m(x) = 0$, is usually adopted and deemed appropriate when outputs represent deviations from a nominal behavior. Given M noisy observations $y = [y_1, \dots, y_M]^T$ at inputs $X = \{x_i\}_{i=1}^M$, where $y_i = h(x_i) + \varepsilon_i$ with $\varepsilon_i \sim \mathcal{N}(0, \sigma_n^2)$, the predictive distribution of h_* at a test point x_* is Gaussian with mean and variance

$$\mathbb{E}[h_*] = k_*^\top (K + \sigma_n^2 I)^{-1} y, \quad (5)$$

$$\text{Var}[h_*] = k(x_*, x_*) - k_*^\top (K + \sigma_n^2 I)^{-1} k_*, \quad (6)$$

where K is the $M \times M$ kernel matrix with entries $K_{ij} = k(x_i, x_j)$, $k_* = [k(x_*, x_1), \dots, k(x_*, x_M)]^\top$ is the vector of covariances between the query point and the training data, and I is the $M \times M$ identity matrix. The choice of the covariance function $k(\cdot, \cdot)$ governs the smoothness and generalization properties of the inferred model.

IV. PROBLEM FORMULATION

We model end-effector motion as first-order autonomous dynamical systems on $\text{SE}(3)$. The robot state is $\xi = (p, q) \in \mathbb{R}^3 \times \mathbb{S}^3$, where $p \in \mathbb{R}^3$ is the Cartesian position and $q \in \mathbb{S}^3$ is the unit quaternion. Position and orientation dynamics are decoupled and learned independently. In this work, the translational motion is learned with the classic LPV-DS framework [3], whereas the rotational motion is learned with a Quaternion-DS formulation [16]. Given the current q and the attractor orientation q_{att} , the Quaternion-DS predicts the next tangent vector as¹

$$\hat{q}_{\text{att}} = \sum_{k=1}^K \gamma_k(q; \Theta_\gamma) A_k \log_{q_{\text{att}}}(q), \quad (7)$$

where K is number of linear systems, $\gamma_k(\cdot)$ are weighting functions and A_k are linear system matrices. Following the

¹Throughout this paper, tangent vectors on the quaternion manifold are denoted in Fraktur typeface; e.g., $\mathbf{q} \in T\mathcal{M}$.

additional steps outlined in Section III of [16], we turn Eq. 7 into a compact representation $\omega = f(q)$, where ω is the angular velocity and $f(\cdot)$ is the mapping of the learned DS.

Throughout this paper, this Quaternion-DS is assumed to be pre-trained and serves as the nominal orientation policy. Our objective is to endow this nominal policy with the adaptability that locally modulates the orientation dynamics to achieve obstacle avoidance behavior. This can be done via a multiplicative modulation matrix:

$$\hat{\omega} = \hat{f}(q) = M(q)f(q), \quad (8)$$

where $M(q)$ is a state-dependent modulation matrix and $\hat{\omega}$ is the modulated angular velocity. The two sections that follow present two constructions of this modulation: (i) a closed-form online operator computed from the obstacle geometry (Section V), and, (ii) for obstacles whose geometry cannot be captured analytically, an offline operator learned from corrective demonstrations (Section VI).

V. ONLINE OBSTACLE AVOIDANCE VIA ORIENTATION MODULATION AND REPULSION

We begin with the online formulation, which can be used whenever the obstacle admits an analytic geometric description. Thus, the modulation can be computed in closed form directly from the obstacle geometry, and combined with its translational counterpart for more dexterous behavior.

A. Problem Setting

Orientation alone occupies no volume, so there is no notion of an obstacle intrinsic to $\text{SO}(3)$, and a rotation can only collide with a physical obstacle through the body points it sweeps through space. Consequently, any orientation-space avoidance problem has to be cast onto a collision condition defined in physical space $\mathbb{R}^3$.

We consider an end-effector whose wrist, i.e., the frame origin, is at a position $p \in \mathbb{R}^3$, with orientation matrix $R \in \text{SO}(3)$. A representative body point is rigidly attached at offset $\ell \in \mathbb{R}^3$ in the end-effector frame; its world position is

$$b(R) = p + R\ell, \quad (9)$$

which, as R varies over $\text{SO}(3)$, $b(R)$ traces the sphere $S_\ell = \{b(R) : R \in \text{SO}(3)\}$ of radius $\|\ell\|$ centered at the wrist p .

We define the obstacle $\mathcal{O}$ as a sphere specified by its center $x_c \in \mathbb{R}^3$ and radius $r > 0$. The collision condition is defined in physical space: the body point collides with the obstacle when it enters $\mathcal{O}$,

$$b(R) \in \mathcal{O} \iff \|b(R) - x_c\| < r. \quad (10)$$

The set of colliding orientations is,

$$\mathcal{C} = \{R \in \text{SO}(3) : b(R) \in \mathcal{O}\}. \quad (11)$$

is therefore not specified directly on the rotation manifold but induced by Eq. (10). The collision set $\mathcal{C}$ is thus a spherical cap on S_ℓ , comprising the orientations that swing the body point toward the obstacle, as shown in Fig. 3.

We assume a nominal DS orientation policy supplies a nominal angular velocity $\omega \in \mathfrak{so}(3) \cong \mathbb{R}^3$ at each instant, encoding the desired rotational motion toward a

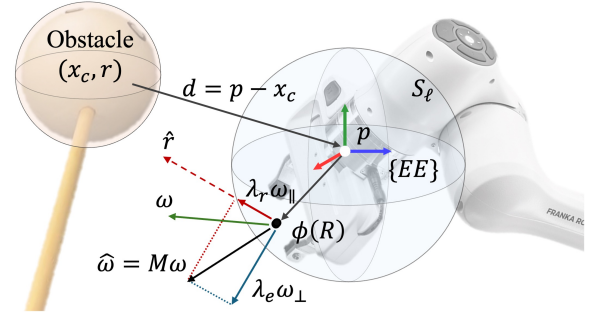


Fig. 3: Schematics of the online modulation for obstacle avoidance.

target orientation in the absence of obstacles. We treat the policy as a black box, as our method requires only its output ω , the current orientation R , and the obstacle geometry (x_c, r) . While our filter is agnostic to the source of ω , in all our experiments the nominal command is supplied by the Quaternion-DS policy i.e., ω is the angular velocity induced by the nominal orientation DS.

The problem is then to design a mapping that converts the nominal command to a modified command,

$$\omega \mapsto \hat{\omega} = g(\omega, R; p, x_c, r), \quad (12)$$

such that: (i) the resulting orientation trajectory never enters the collision set $\mathcal{C}$; and (ii) the mapping recovers the nominal command, $\hat{\omega} = \omega$, away from the obstacle.

B. Method

Let the obstacle function on $\text{SO}(3)$ be the composition of the Euclidean obstacle field with the rigid-body point map,

$$\Gamma_{\mathbb{R}^3}(x) = \frac{\|x - x_c\|^2}{r^2}$$

denote the physical space obstacle field on $\mathbb{R}^3$, and let

$$\phi : \text{SO}(3) \rightarrow \mathbb{R}^3, \quad \phi(R) = p + R\ell,$$

map an orientation to the world position of the body point. Note that ϕ is precisely the body-point in Eq. (9) but we introduce the separate symbol ϕ to emphasize its role as the smooth map $\text{SO}(3) \rightarrow \mathbb{R}^3$ along which the Euclidean obstacle field is pulled back. The orientation obstacle function is the pullback of $\Gamma_{\mathbb{R}^3}$ along ϕ :

$$\Gamma(R) = (\phi^* \Gamma_{\mathbb{R}^3})(R) = \Gamma_{\mathbb{R}^3}(\phi(R)) = \frac{\|p + R\ell - x_c\|^2}{r^2}. \quad (13)$$

This construction transfers the collision condition from $\mathbb{R}^3$ to the rotation manifold. Equivalently, the collision set

$$\mathcal{C} = \{R \in \text{SO}(3) : \Gamma(R) < 1\} \quad (14)$$

is the preimage of the obstacle under ϕ . Writing $d = p - x_c$ for the wrist-to-obstacle displacement and using $\|R\ell\| = \|\ell\|$, the obstacle function is rewritten and expanded:

$$\Gamma(R) = \frac{\|d + R\ell\|^2}{r^2} = \frac{\|d\|^2 + \|\ell\|^2 + 2d^\top R\ell}{r^2} \quad (15)$$

Note that Γ is bounded above since S_ℓ is compact. Hence, there is no ∞ to recede to, as in Euclidean obstacle fields.

a) *The reference direction:* The reference direction is the axis about which an infinitesimal rotation increases the obstacle function Γ most rapidly. Equivalently, it is the normalized gradient of Γ on the tangent space of $\text{SO}(3)$.

This gradient is obtained by pulling the Euclidean gradient of the obstacle field back through the body-point map.

For an infinitesimal rotation perturbation $\delta\theta$, first-order displacement of the body-fixed offset ℓ is therefore obtained by rotating $R\ell$ about the small axis $\delta\theta$:

$$\delta(R\ell) = \delta\theta \times (R\ell) = -[R\ell]_{\times} \delta\theta. \quad (16)$$

Since $b(R) = p + R\ell$ and p is constant,

$$\delta b = \delta(R\ell) = -[R\ell]_{\times} \delta\theta. \quad (17)$$

Using the Jacobian definition as the linear map relating perturbations $\delta b = J_b \delta\theta$, we identify $J_b = -[R\ell]_{\times}$. We then derive the Euclidean gradient, as the obstacle field Eq. 15 has the following gradient and is evaluated at $x = \phi(R)$:

$$g_x := \nabla_x \Gamma_{\mathbb{R}^3}(\phi(R)) = \frac{2}{r^2} (d + R\ell). \quad (18)$$

To pullback the Jacobian to the orientation tangent space, we apply the chain rule,

$$\nabla_{\text{SO}(3)} \Gamma = J_b^{\top} g_x = \frac{2}{r^2} (R\ell) \times d, \quad (19)$$

and the resulting reference direction is then the normalized gradient as follows:

$$\hat{r}(R) = \frac{(R\ell) \times d}{\|(R\ell) \times d\|}. \quad (20)$$

For any angular velocity ω ,

$$\dot{\Gamma} = \langle \nabla_{\text{SO}(3)} \Gamma, \omega \rangle. \quad (21)$$

Among all unit-norm angular velocities, $\dot{\Gamma}$ is maximized when $\omega = \hat{r}(R)$. Hence $\hat{r}(R)$ is the instantaneous rotation axis that increases Γ most rapidly and rotates the body point away from the obstacle center at the max. first-order rate.

b) Modulation operator: Given the nominal angular velocity ω , we decompose it into a component aligned with the reference direction $\hat{r}$ and a component orthogonal to it:

$$\omega = \omega_{\parallel} + \omega_{\perp}, \quad \omega_{\parallel} = (\hat{r}^{\top} \omega) \hat{r}, \quad \omega_{\perp} = (I - \hat{r} \hat{r}^{\top}) \omega. \quad (22)$$

The modulation acts independently on the two components, with $\lambda_r = 1 - \Gamma^{-1}$ and $\lambda_e = 1 + \Gamma^{-1}$:

$$M(R) \omega = \lambda_r \omega_{\parallel} + \lambda_e \omega_{\perp}. \quad (23)$$

Because the orthogonal response is chosen to be isotropic, Eq. 23 admits the following compact matrix representation,

$$M(R) = \lambda_r \hat{r} \hat{r}^{\top} + \lambda_e (I - \hat{r} \hat{r}^{\top}) = \lambda_e I + (\lambda_r - \lambda_e) \hat{r} \hat{r}^{\top}. \quad (24)$$

Through eigendecomposition we can see that $\hat{r}$ is an eigenvector with eigenvalue λ_r , whereas every vector in the orthogonal subspace is an eigenvector with eigenvalue λ_e . Thus, the modulation scales motion differently along the reference direction and within its orthogonal complement.

Substituting $\lambda_r - \lambda_e = -2\Gamma^{-1}$ yields an implementation-friendly expression that avoids explicitly forming $M(R)$:

$$M(R) \omega = (1 + \Gamma^{-1}) \omega - 2\Gamma^{-1} (\hat{r}^{\top} \omega) \hat{r}. \quad (25)$$

Far from the obstacle ($\Gamma \rightarrow \infty$), both gains converge to one and the nominal velocity is recovered. On the obstacle boundary ($\Gamma = 1$), the reference direction component is completely suppressed ($\lambda_r = 0$), preventing further motion toward the obstacle. Inside the obstacle ($\Gamma < 1$), $\lambda_r < 0$, reversing the aligned component and driving the state outward. Meanwhile, $\lambda_e > 1$ amplifies motion in the

orthogonal subspace, encouraging circumvention rather than penetration. This behavior mimics the standard obstacle-avoidance modulation in Euclidean space.

c) Repulsion operator: While the previous modulation is multiplicative, we include an additive term that will work for stationary configurations when angular velocity vanishes $\omega = 0$. We therefore introduce a repulsion term

$$\omega_{\text{rep}}(R) = k \Gamma^{-1} \hat{r}, \quad k > 0. \quad (26)$$

The repulsion acts along the reference direction $\hat{r}$, i.e. the steepest ascent direction of Γ , and therefore increases the body point to obstacle distance even when the nominal command is zero. Its magnitude grows as the obstacle is approached ($\Gamma \rightarrow 1$) and decays to zero in free space, as in the classical obstacle-avoidance modulation.

d) Complete filter: Combining modulation and repulsion yields the orientation safety filter

$$\hat{\omega} = M(R) \omega + \omega_{\text{rep}}(R). \quad (27)$$

The modulation term, $M(R)$, reshapes a nonzero nominal command to steer the flow around the obstacle, while the repulsion, $\omega_{\text{rep}}(R)$, provides avoidance when the nominal command vanishes. The modulated behavior reduces to the identity in free space. As $\Gamma \rightarrow \infty$, we have $M(R) \rightarrow I$ and $\omega_{\text{rep}}(R) \rightarrow 0$, yielding $\hat{\omega} \rightarrow \omega$ and recovering the nominal generator unchanged, offering the same smooth yet reactive obstacle-avoidance behavior as the standard modulation approaches. Furthermore, we can also combine the orientation modulation with its translational counterpart.

If the orientation modulation is defined at the wrist which rotates the end-effector to avoid obstacles, we can also translate the wrist itself if needed. Specifically, if a linear DS defined as $\dot{x} = g(x)$, we can introduce the translational modulation as follows, $\hat{x} = \hat{g}(x) = M_g(x)g(x)$, where $M_g(x)$ is the modulation matrix associated with the position x . The construction of $M_g(\cdot)$ follows the original formulation [9], which similarly shrinks the component of the translational DS toward the obstacle and amplifies the tangential ones. We note that since $\hat{f}(q)$ modulates the orientation and $\hat{g}(x)$ modulates the translation, both modulations can occur concurrently, and hence no scheduler or switching mechanism is needed.

VI. LOCALLY MODULATED QUATERNION DS

The online filter of the previous section hinges entirely on the availability of an analytic obstacle field $\Gamma_{\mathbb{R}^3}$. Yet, for many practical obstacles their geometry may be complex or partially occluded and no reliable primitive can be fitted, e.g., narrow passages or irregularly shaped objects viewed from limited camera angles, in which case the reference direction $\hat{r}(R)$ can no longer be computed. In this regime, the desired avoidance behavior can still be specified by the most general oracle of all: a demonstration. A human teacher who guides the robot around an obstacle implicitly encodes both the obstacle geometry and the preferred avoidance style, without ever requiring an explicit geometric model. This section therefore develops an offline method that learns the orientation modulation from such corrective demonstrations.

Extending LMDS to orientation policies, however, is significantly more challenging due to the non-Euclidean geometry of $\text{SO}(3)$. A naive additive correction does not respect the geometry of the quaternion manifold and, if interpreted directly in quaternion coordinates, may violate the unit-norm constraint of $\mathbb{S}^3$. We therefore propose a two-stage framework. During the *training* phase, geometric modulation parameters are extracted from corrective demonstrations and used to train a GP model. During the *inference* phase, the learned GP predicts the state-dependent multiplicative modulation, which is applied to reshape the nominal Quaternion-DS for desired obstacle avoidance behavior.

A. Geometric Parameter Extraction and GP Training

We first define the modulation training set as M pairs of unit quaternions $\{q_{\text{in}}^i, \hat{q}_{\text{out}}^i\}_{i=1}^M$, where q_{in}^i denotes the current orientation and $\hat{q}_{\text{out}}^i$ is the desired target orientation under modulation. Given a nominal DS $f(\cdot)$, our objective is to learn a modulated DS $\hat{f}(\cdot)$, such that the modulated response aligns with the behavior specified in the training set: $\hat{f}(q_{\text{in}}^i) \approx \hat{q}_{\text{out}}^i$. We first compute the original output denoted by $q_{\text{out}}^i := f(q_{\text{in}}^i)$ ². These are computed by evaluating the nominal DS at all inputs q_{in}^i , $i = 1, \dots, M$ in the modulation training set. To extract the geometric parameters θ_i corresponding to each pair $\{q_{\text{out}}^i, \hat{q}_{\text{out}}^i\}$, we first need to transform each unit quaternion into the appropriate form by recognizing the Riemannian structure of $\mathbb{S}^3$ and leveraging the operators outlined in Sec. III-A. We begin by projecting each output quaternion onto the tangent plane of its corresponding input via the logarithmic map defined in Eq. 2:

$$q_{\text{in}}^{\text{out}} = \log_{q_{\text{in}}}(q_{\text{out}}); \quad \hat{q}_{\text{in}}^{\text{out}} = \log_{q_{\text{in}}}(\hat{q}_{\text{out}}), \quad (28)$$

where we omit the index i for notational simplicity hereinafter. We then apply parallel transport as in Eq. 3 to group each resulting output into a common tangent plane defined by the attractor q_{att} :

$$q_{\text{att}}^{\text{out}} = \Gamma_{q_{\text{in}} \rightarrow q_{\text{att}}}(q_{\text{in}}^{\text{out}}); \quad \hat{q}_{\text{att}}^{\text{out}} = \Gamma_{q_{\text{in}} \rightarrow q_{\text{att}}}(\hat{q}_{\text{in}}^{\text{out}}). \quad (29)$$

This tangent space is a 3D hyperplane embedded in $\mathbb{R}^4$, for which we construct an orthonormal basis $\mathbf{B} \in \mathbb{R}^{4 \times 3}$ spanning the null space of q_{att} :

$$\mathbf{B} = [b_1, b_2, b_3], \quad \text{s.t. } b_i^\top q_{\text{att}} = 0, \quad \forall i \in 1, 2, 3. \quad (30)$$

We then apply the coordinate transformation by projecting the tangent vectors onto the basis $\mathbf{B}$ using a least-squares approach:

$$u = \mathbf{B}^\dagger q_{\text{att}}^{\text{out}}, \quad \hat{u} = \mathbf{B}^\dagger \hat{q}_{\text{att}}^{\text{out}}, \quad (31)$$

where $\mathbf{B}^\dagger$ denotes the Moore-Penrose pseudoinverse, yielding the 3D reduced representations $u, \hat{u} \in \mathbb{R}^3$. We can then extract the geometric parameters $\theta_i = [\kappa, \mu, \phi]$ from the pairs $\{u, \hat{u}\}$ as follows:

$$\kappa = \frac{|\hat{u}|}{|u|} - 1, \quad \phi = \arccos\left(\frac{u^\top \hat{u}}{|u||\hat{u}|}\right), \quad \mu = \frac{u \times \hat{u}}{|u \times \hat{u}|}. \quad (32)$$

²With a slight abuse of notation, we use $f(\cdot)$ here to denote the nominal orientation DS mapping the current orientation to the next orientation, i.e., the orientation reached by integrating the angular velocity $\omega = f(q_{\text{in}})$ of Section IV over one control step Δt .

Algorithm 1 Computing modulated target orientation

Require: Current orientation q_{in}^t , attractor q_{att} , trained $\mathcal{GP}$
Ensure: Modulated target orientation $\hat{q}^{\text{out}}$

- 1: **for** each time step t **do**
- 2: Compute input feature: $x_t = \log_{q_{\text{att}}}(q_{\text{in}}^t)$
- 3: Compute nominal output: $q_{\text{out}}^t = f^t(q_{\text{in}}^t)$
- 4: Apply Eqs. 28–31 to get u_t
- 5: Apply GPR: $(\hat{\theta}_t, \hat{\sigma}_t) \leftarrow \mathcal{GP}(x_t)$
- 6: Construct modulation matrix: $M_t = (1 + \hat{\kappa}_t) R(\hat{\phi}_t \hat{\mu}_t)$
- 7: Apply modulation: $\hat{u}_t = M_t u_t$
- 8: Transform coordinates: $\hat{q}_{\text{att}}^{\text{out}} = \mathbf{B} \hat{u}_t$
- 9: Parallel transport: $\hat{q}_{\text{in}}^{\text{out}} = \Gamma_{q_{\text{att}} \rightarrow q_{\text{in}}^t}(\hat{q}_{\text{att}}^{\text{out}})$
- 10: Output retrieval: $\hat{q}^{\text{out}} = \exp_{q_{\text{in}}^t}(\hat{q}_{\text{in}}^{\text{out}})$
- 11: **end for**
- 12: **return** Modulated orientation $\hat{q}^{\text{out}}$

To achieve true geometric smoothness directly on the unit quaternion manifold $\mathbb{S}^3$, the covariance function $k(q, q')$ can be formulated using the geodesic distance $d(q, q') = \arccos(q^\top q')$. This yields a geometry-aware squared-exponential kernel:

$$k(q, q') = \exp\left(-\frac{\arccos(q^\top q')^2}{2\ell^2}\right) \quad (33)$$

This formulation ensures that the control commands (tangent vectors) rotate smoothly as the robot moves through the orientation space, while effectively filtering out high-frequency noise in the demonstrations through the noise variance term $\sigma_n^2 \mathbf{I}$. Since the modulation parameters θ are multidimensional, we employ four independent GPs to learn each component separately, specifically for κ , $\mu_x \phi$, $\mu_y \phi$ and $\mu_z \phi$, sharing the same set of hyperparameters for efficiency.

B. Inference and Modulation at Runtime

At each control step t , the nominal DS provides a desired orientation q_{out}^t . To apply the learned modulation, we first acquire the reduced representation u_t , following the steps from Eq. 28 to Eq. 31. We then query the learned GP regressor using the feature input $x_t = \log_{q_{\text{att}}}(q_{\text{in}}^t)$ to retrieve the predicted modulation parameters $\hat{\theta} = [\hat{\kappa}, \hat{\mu}, \hat{\phi}]$. From these regressed parameters, we construct the modulation matrix $M_t(R)$ and multiply it by u_t to obtain the modulated reduced representation $\hat{u}_t$. To recover the unit quaternion representation, we need to perform a series of reverse operations. The complete pipeline is outlined in Algorithm 1.

C. Stability Analysis

The stability of the proposed framework builds upon two existing theoretical results. First, the nominal Quaternion-DS was shown in [16] to be globally asymptotically stable (GAS) in the tangent space at the attractor. Second, the LMDS framework [17] establishes sufficient conditions under which a multiplicative modulation preserves the local asymptotic stability of the nominal attractor. In particular, if the modulation satisfies the conditions derived in [17], which are that $M(\cdot)$ should be (i) locally active within a compact region that does not include the attractor and (ii) full rank, then the original equilibrium remains locally asymptotically stable.

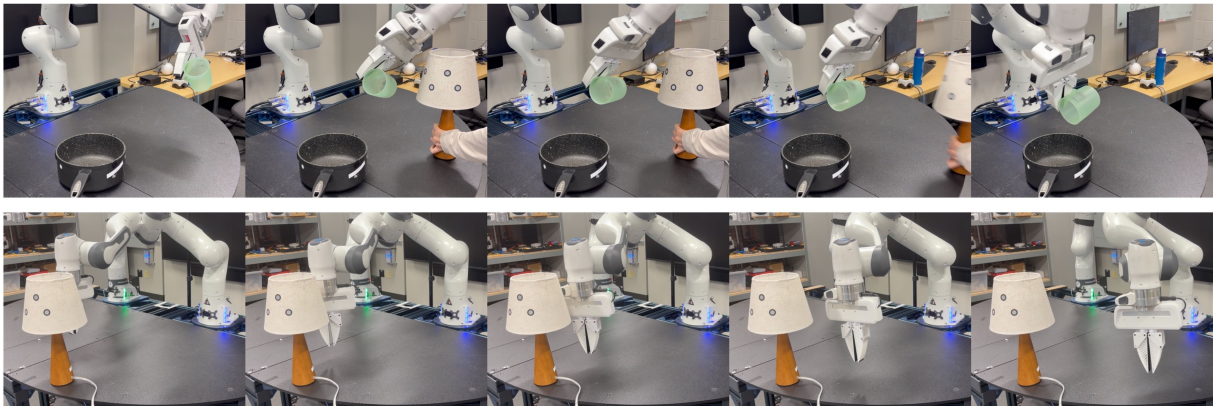


Fig. 4: In the pouring task (top row), the robot avoids the lamp by tilting its end effector; during the traversing task (bottom row) the robot avoids the lamp by only turning its end effector in yaw angle; both while staying on its nominal positional trajectory.

Our proposed framework performs modulation entirely in the quaternion tangent space, where the nominal Quaternion-DS is defined. Since the proposed coordinate transformations map between tangent spaces without altering the underlying manifold structure, the LMDS stability analysis remains applicable in the transformed coordinates. Since our modulation operators defined in Section V and VI are full-rank and locally active, **by construction**, our modulation approach on orientation space preserves the local stability properties of the nominal DS, just as its translational counterpart.

VII. EVALUATION

We now evaluate both online and offline methods on a 7-DoF Franka Panda across three obstacle-avoidance scenarios: (i) pouring, (ii) cross-table traversal, and (iii) book placing under a shelf. For each, the baseline policy or the nominal DS was trained from five demonstrations provided by human experts who guided the robot to accomplish the task in an obstacle-free environment through kinesthetic teaching. The trained policy is a SE(3) LPV-DS [16] that encodes both position and orientation planning and control.

A. Online Modulation

We first evaluate the online mode with obstacles that admit a spherical approximation, incorporating a tracking system to actively estimate the pose of both stationary and dynamic objects. For the pouring task, we actively track the obstacle, which is a lamp with reflective stickers for localizations. Shortly after the robot starts executing its closed-loop policy, we introduced the obstacle by engaging with the end effector. As shown in the top row of Fig. 4, the robot’s end effector starts tilting towards the robot base to avoid the approaching obstacle, while its position was not affected and remained on track. After we removed the lamp, the robot then resumed its nominal orientation behavior to complete the rest of the task.

For cross-table traversal task, we used the same setup as the previous example except we have the lamp as a stationary obstacle. In this case, the robot starts with a configuration that would directly collide with the obstacle if no modulation is performed. As shown in the bottom row of Fig. 4, the online method successfully recognized the proper response by turning the end-effector in its yaw angle to avoid the

lamp. Similarly, the positional path of the nominal trajectory remained unchanged, and all the avoidance was achieved by modulating in orientation space.

Implementation details: The body point in Eq. 9 has to be manually measured and defined. The placement of this body point could result in different avoidance behavior. We define multiple body points on the end effectors including gripper tip and the left and right lateral sides of the gripper body. And the reference direction was selected only with respect to the body point that has the lowest Γ value.

B. Offline Modulation

We next evaluate the offline mode in the regime the online operator cannot handle, where the obstacle geometry is not captured by a simple analytic primitive. The modulation training set is collected by re-teaching the same task with the obstacle present. The human teacher guides the end-effector through a modified orientation trajectory that avoids the obstacle while preserving the position objective. The geometric modulation parameters $(\kappa, \phi\mu)$ are extracted and fed into the training framework outlined in Sec. VI. At inference the modulated policy is queried in closed loop following Algorithm 1 and tracked by the robot’s built-in Cartesian impedance controller. We use the pouring task as an illustrative example. Because the original DS policy is inherently rigid and unaware of the environment, executing the baseline motion would result in a collision as shown in the top row of Fig. 5. For the modulated policy, the GP regression model successfully learns the mapping from the local orientation states to the required modulation parameters. At runtime, the modulated policy correctly anticipates the obstacle and reshapes the vector field to avoid it. Once the obstacle is cleared, the system safely falls back to the convergence-guaranteed nominal DS as shown in the bottom row of Fig. 5. *For the other two tasks:* passage traversal and book placing, we show similar results in the video where the baseline fails. Given corrective training data provided by a human, we avoid the need to re-collect demonstrations and retrain the entire policy, while still being able to achieve the desired obstacle avoidance behavior. We note that although correction data is needed for offline approach, we are able to inject more flexible avoidance behavior as the training data

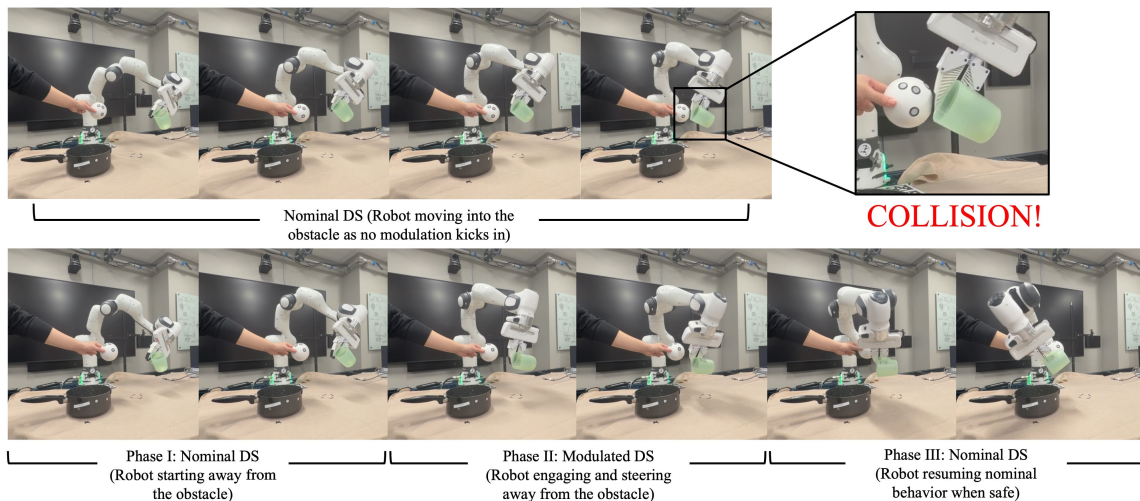


Fig. 5: Comparisons of baseline policy with the proposed method, where the top row shows that the robot moved into the obstacle and caused collision as no modulation kicks in to steer the robot away from the obstacle in the orientation space, and the bottom row shows that robot switched between nominal and modulated behavior as the transition was entirely governed by the GP.

is directly from humans who can explicitly introduce more expressive motion.

VIII. CONCLUSION

This paper introduced a framework for reactive obstacle avoidance defined explicitly in the $SO(3)$ orientation space of a DS motion policy. Our primary, online formulation pulls a Euclidean obstacle field back onto the rotation manifold via the rigid-body point map and computes a closed-form modulation and repulsion response on the fly, requiring no data collection. For obstacles whose geometry cannot be captured analytically, we introduced a complementary offline formulation that extends the LMDS framework to the quaternion manifold, learning the modulation from a handful of corrective kinesthetic demonstrations. Instead of modulating position, which is the more common route to obstacle avoidance, our method opens up a new and more expressive avenue for obstacle avoidance in orientation space.

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